

Question 1 (Part a, b, c)

①

Part (a) For a 3×4 matrix in form of $m \times n$, $m < n$
Let us assume below equations

$$\begin{aligned}x_1 + 2x_2 + x_3 + x_4 &= 5 \\2x_1 + 5x_2 + 3x_3 + 2x_4 &= 12 \\x_1 + x_2 + 2x_3 + x_4 &= 6\end{aligned}$$

Writing in matrix form

$$A = \begin{bmatrix} 1 & 2 & 1 & 1 \\ 2 & 5 & 3 & 2 \\ 1 & 1 & 2 & 1 \end{bmatrix}, b = \begin{bmatrix} 5 \\ 12 \\ 6 \end{bmatrix}$$

Constructing augmented Matrix

$$[A|b] = \left[\begin{array}{cccc|c} 1 & 2 & 1 & 1 & 5 \\ 2 & 5 & 3 & 2 & 12 \\ 1 & 1 & 2 & 1 & 6 \end{array} \right]$$

Converting above matrix to REF form
 $\Rightarrow R_2 = R_2 - 2R_1$

$$[A|b] = \left[\begin{array}{cccc|c} 1 & 2 & 1 & 1 & 5 \\ 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & 2 & 1 & 6 \end{array} \right]$$

$$\Rightarrow R_3 = R_3 - R_1$$

②

$$[A|b] = \left[\begin{array}{cccc|c} \underline{1} & 2 & 1 & 1 & 5 \\ 0 & \underline{1} & 1 & 0 & 2 \\ 0 & -1 & 1 & 0 & 1 \end{array} \right]$$

$$a_{22} = 1$$

So eliminate below it.

$$\Rightarrow R_3 = R_3 + R_2 \text{ gives}$$

$$[A|b] = \left[\begin{array}{cccc|c} \underline{1} & 2 & 1 & 1 & 5 \\ 0 & \underline{1} & 1 & 0 & 2 \\ 0 & 0 & \underline{2} & 0 & 3 \end{array} \right]$$

Final REF

• Converting now to RREF form

so now we need to make every pivot to 1 & entries above it as 0.

$$\Rightarrow R_3 = \frac{1}{2} R_3$$

$$[A|b] = \left[\begin{array}{cccc|c} \underline{1} & 2 & 1 & 1 & 5 \\ 0 & \underline{1} & 1 & 0 & 2 \\ 0 & 0 & \underline{1} & 0 & 3/2 \end{array} \right]$$

Now remove the elements above pivot in row 3

$$\Rightarrow R_2 = R_2 - R_3$$

$$\Rightarrow R_1 = R_1 - R_3$$

$$[A|b] = \left[\begin{array}{cccc|c} \underline{1} & 2 & 1 & 1 & 5 \\ 0 & \underline{1} & 0 & 0 & 1/2 \\ 0 & 0 & \underline{1} & 0 & 3/2 \end{array} \right]$$

$$[A|b] = \left[\begin{array}{cccc|c} \underline{1} & 2 & 0 & 1 & 7/2 \\ 0 & \underline{1} & 0 & 0 & 1/2 \\ 0 & 0 & \underline{1} & 0 & 3/2 \end{array} \right]$$

Now:

$$\Rightarrow R_1 = R_1 - 2R_2$$

$$[A|b] = \left[\begin{array}{cccc|c} 1 & 0 & 0 & 1 & 5/2 \\ 0 & 1 & 0 & 0 & 1/2 \\ 0 & 0 & 1 & 0 & 3/2 \end{array} \right]$$

So:

$$REF = \left[\begin{array}{cccc|c} 1 & 2 & 1 & 1 & 5 \\ 0 & 1 & 1 & 1 & 2 \\ 0 & 0 & 2 & 1 & 3 \end{array} \right]$$

$$RREF = \left[\begin{array}{cccc|c} 1 & 0 & 0 & 1 & 5/2 \\ 0 & 1 & 0 & 0 & 1/2 \\ 0 & 0 & 1 & 0 & 3/2 \end{array} \right]$$

Final answer
for Q1(a)

Q1) Part(b) Pivot columns, Non-Pivot columns,
Particular solution & Solutions of $Ax=0$

Pivot columns are columns containing leading 1s

So Pivot columns are 1, 2, 3, Non-pivot column is

4

Therefore x_1, x_2, x_3 are pivot variables
& x_4 is the free variable.

Writing equations from RREF

Row 1: $x_1 + x_4 = 5/2$

Row 2: $x_2 = 1/2$

Row 3: $x_3 = 3/2$

(4)

Pivot variables in terms of free variables: $x_4 = t$
 $t \in \mathbb{R}$

$$x_1 = \frac{5}{2} - t$$

$$x_2 = 1/2$$

$$x_3 = 3/2$$

$$x_4 = t$$

$$\text{hence } x = \begin{bmatrix} 5/2 - t \\ 1/2 \\ 3/2 \\ t \end{bmatrix}$$

Writing general solution $Ax = b$

$$x = \begin{bmatrix} 5/2 \\ 1/2 \\ 3/2 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix} \quad t \in \mathbb{R}$$

So, Particular solution x_p

$$x_p = \begin{bmatrix} 5/2 \\ 1/2 \\ 3/2 \\ 0 \end{bmatrix}$$

even if we set $x_4 = 0$ (free variable)
 we can get x_p (PTO)

To get Particular solution setting free variable ⁽⁵⁾
 $x_4 = 0$

So: $x_1 = 5/2$, $x_2 = 1/2$, $x_3 = 3/2$, $x_4 = 0$

$$x_p = \begin{bmatrix} 5/2 \\ 1/2 \\ 3/2 \\ 0 \end{bmatrix}$$

← Particular
solution

Now: homogenous system $Ax = 0$] Null space

$$x_1 + x_4 = 0$$

$$x_2 = 0$$

$$x_3 = 0$$

Let $x_4 = t$, $t \in \mathbb{R}$

then $x_1 = -t$

hence, $x = \begin{bmatrix} -t \\ 0 \\ 0 \\ t \end{bmatrix}$

$$x = t \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

} Null space
solution